Analytical Verification and Targeted FEM Diagnostic: Rev0 and Rev1-S

EEE4022S AK-07 · Oct 22, 2026 · @Olwethu

Summary

The magnetic field of the original Rev0 and Rev1-S designs is confirmed independently. The roughly 5 % difference between the field-only models and the detailed-copper FEM comes from how the conductors are represented, not from the field.

- **Independent analytical model.** No Maxwell input. It predicts K = 19.50 mN·m/A (Rev0) and 41.65 mN·m/A (Rev1-S). This agrees with the FEM field map within 0.3 % per stator, including the different steel surroundings of Rev1-S S2.
- **Against the detailed-copper FEM.** The analytical K reads +5.2 % (Rev0) and +5.45 % (Rev1-S) high. Per board, the gap grows from +1.4 % at the board nearest the magnets to +10 % at the farthest.
- **Targeted FEM diagnostic** (two HPC jobs, board 1 and board 5):
  - Within each detailed model, Maxwell's own magnet field, integrated along the same trace paths, reproduces the field map within 0.6 %.
  - Maxwell's copper and rotor torques agree within 0.4 %.
  - The remaining gap (+1.3 % at board 1, +9.7 % at board 5) lies in the conductor representation: centre-line trace paths with DC-network current shares, against Maxwell's solid copper. An offline check confirms it: with the DC current solved as 2D sheets in the real copper islands, the analytical model reproduces Maxwell's detailed K within 0.02 % at both positions.
- **Consequence for reported values.** The detailed-copper K (18.53 and 39.50 mN·m/A) remains the reference. The analytical model is an independent confirmation of the field, phase, harmonics and S2 effect. With centre-line conductors it reads +5 % on K. With the current solved in the real copper it predicts K = 18.54 and 39.55 mN·m/A, within 0.03 % and 0.13 % of the FEM.

Throughout, values are labelled **analytical**, **FEM** or **measured**. No new experimental results are used; bench testing remains postponed.

Background

Since 8 October, two FEM routes to the torque constant have disagreed by about 5 % per stator. That gap was explained only by assumption.

- **Detailed-copper FEM.** Each board modelled with its real copper, solved one board at a time at its axial position. Board sum 18.54 mN·m/A (Rev0), confirmed by the full-motor HPC solve, 18.534.
- **Field map.** Magnets and steel only. The field is sampled along the copper trace centre-lines and the Lorentz force summed: 19.50 mN·m/A.

The field map reads 1.4 % high at board 1 (1.0 mm from the magnets), rising to 10 % at board 5 (5.08 mm). Mesh, region size and shaft geometry were ruled out at the time (each ⩽ 0.1 %). The gap was then attributed to the centre-line (filament) approximation without a direct test.

Two questions follow from that:

1. Is the field itself right? A field error would affect the back-EMF and every rotating result, not just K.
2. Which K should the report treat as the reference?

This document answers both. It uses an independent analytical model, which shares no code or field data with Maxwell, and a targeted two-position FEM diagnostic.

All geometry is the **original design**:

- 86 mm OD / 19 mm bore / 2 mm steel backplates;
- 16 N35 magnets, 20 × 6 × 2 mm, on r = 28 mm;
- 1.0 mm magnet-to-PCB gap and 1.0 mm R1-steel-to-S2 clearance;
- 5 × 1.02032 mm boards per stator.

The manufactured prototype geometry is not used.

Independent analytical model

The analytical model computes field, torque and back-EMF from geometry, layout copper paths and magnet data alone. No Maxwell field, torque or flux enters it.

Inputs

- **Copper paths.** Extracted from the ODB++ manufacturing export, independent of Ansys. Every track centre-line on L1–L6 and every via step is included, with its share of 1 A from a DC nodal solve of the copper network. The parallel L2/L3 and L4/L5 copper is handled by that solve.
- **Magnets.** Uniform axial magnetisation alternating N/S, with Br = 1.23 T and μr = 1.0998. These are the N35 material values of the original design, used as material data, not as a result.
- **Steel.** Infinitely permeable and unsaturated. The FEM steel peaks at 0.84–0.93 T.

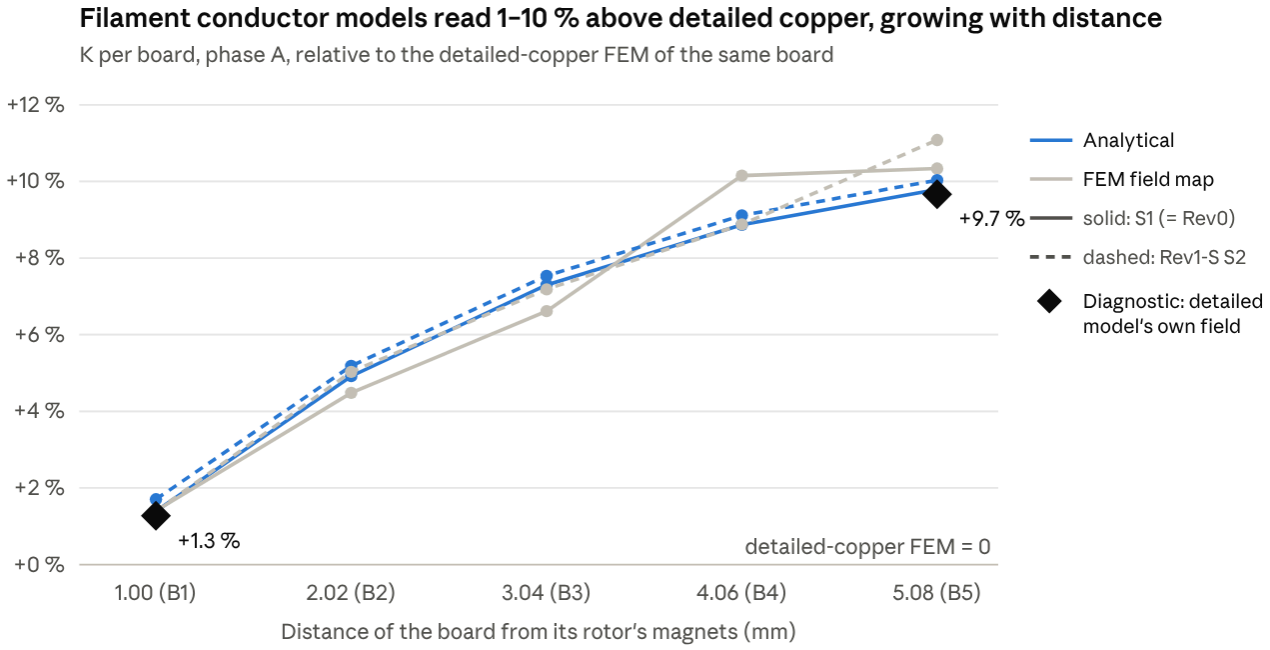
Method

1. **Magnets.** Each magnet is represented by two uniformly charged sheets on its faces. The field of a charged rectangle is closed-form; it was checked against numerical quadrature to five significant figures.
2. **Steel, by images.**
   - Rev0 and Rev1-S S1: one plane (the backplate face). A magnet on its plate becomes a magnet of doubled thickness.
   - **Rev1-S S2:** treated explicitly as a different environment. S2 lies between R1′s bare steel below and R2′s magnets and plate above. Two parallel planes give an infinite image series, truncated at n = 4 (n = 2 and n = 6 change K by 0.01 %). R1′s own magnets are screened by R1′s plate.
3. **Recoil permeability.** The effective magnetisation is $M_{eff} = M / (1 + (\mu_r - 1) N_z)$, with N_z the volume-average demagnetising factor of the image-equivalent magnet: 0.536 for a magnet on one plate, 0.515 for R2 between two plates. This choice was fixed before any comparison was run.
4. **Torque.** The Lorentz torque is summed over every track piece and via step: $T = r \times (I\, dl \times B)$. By reciprocity, $k(\theta) = d\lambda/d\theta = -T/I$. The rotor is turned through one electrical period at 16 angles. The fundamental K, the phase ψ and the 3rd, 5th and 7th harmonics are taken from k(θ).
5. **Back-EMF.** $e = \omega_m \cdot k(\theta)$, so $E_{ph,rms} = K\, \omega_m / \sqrt{2}$. Triplen harmonics cancel in the line voltage.

The torque constant follows the report′s definition: $T = 1.5 \cdot K \cdot \hat{I}$.

Analytical results against FEM

The analytical model matches the FEM field map within 0.3 % per stator and phase. It reads about 5 % above the detailed-copper FEM, with the same per-board trend as the field map.



Analytical model (22 Oct); FEM field map FIELDMAP_REV0_P20 / REV1S_P20; detailed copper REV0_DOE_D*_P20 and REV1S_S2_J*_P20; diagnostic jobs 1517602 and 1517777

The analytical model and the field map lie on top of each other at every board of both stators. Both diverge from the detailed copper by the same distance-growing amount, and the diagnostic points (the detailed model's own field) fall on the same curve.

Quantity	Analytical	FEM field map	FEM detailed copper	Analytical vs field map	Analytical vs detailed
Rev0 K, phase A / B / C (mN·m/A)	19.50 / 19.50 / 19.50	19.50 / 19.45 / 19.49	18.534 (full motor)	0.00 / +0.25 / +0.03 %	+5.2 %
Rev1-S S1 K, phase A (mN·m/A)	19.50	19.50	18.54	−0.02 %	+5.2 %
Rev1-S S2 K, phase A (mN·m/A)	22.16	22.14	20.96	+0.10 %	+5.7 %
Rev1-S K in series, phase A (mN·m/A)	41.65	41.64	39.50 (board sum)	+0.04 %	+5.45 %
S2 / S1 ratio	1.136	1.135	1.131	+0.1 %	+0.5 %
ψ, phase A (deg elec)	−29.98	−30.06	−30.18	+0.08°	+0.20°
Rev0 phase-EMF harmonics H3 / H5 / H7	5.8 / 0.64 / 0.14 %	—	5.5 / 0.7 / 0.1 % (rotating FEM)	—	—
Line back-EMF at 1000 rpm, Rev0 / Rev1-S (V rms)	2.50 / 5.34	—	2.38 / 5.07	—	+5.1 / +5.4 %

Rev1-S S2. R1's bare steel below S2 raises the board next to it from 2.005 to 2.857 mN·m/A. That is +42 % over the same board position in S1, and the field map shows the same (+43 %).

Point check of the field. At the field map's 20 Maxwell probe points (all five board levels, 0°), the rms vector difference is 0.009 T, against an rms field of 0.088 T. The largest differences sit near magnet edges, where Maxwell's 0.8 mm mesh is coarsest relative to the field gradient.

Sensitivity of the analytical K

Case	Rev0	Rev1-S
Conductors averaged over the 0.5 mm × 35 µm cross-section	−0.07 %	−0.07 %
No recoil correction (µr = 1)	+5.3 %	+5.2 %
No steel (air only)	−37 %	−41 %
32 rotor angles instead of 16	+0.01 %	—
Image series n = 2 / n = 6 (S2)	—	−0.01 / 0.00 %
Br = 1.21 T / 1.17 T (N35 range; K ∝ Br)	−1.6 / −4.9 %	−1.6 / −4.9 %

The cross-section result matters here. Spreading each track's current over its real width and thickness changes K by only 0.07 %, so the centre-line approximation does not explain the 5 % gap.

Targeted FEM diagnostic: design

The diagnostic re-solves the archived detailed one-board model at two positions. It separates three possible causes of the gap: the field, Maxwell's force calculation, and the conductor representation.

- **Positions.**
  - Board 1, 1.000 mm from the magnets: the control, where the gap is 1.4 %.
  - Board 5, 5.081 mm: where the gap is 10 %.
- **Settings.** Identical to the archived detailed runs (REV0_DOE_D100_P20 and D508_P20):
  - one board, phase A copper and the rotor, with 20 mm region padding;
  - 4 adaptive passes at 20 % refinement, gap mesh = distance / 2;
  - rotor angles 0, 5.625, 11.25 and 16.875° mechanical.

- **Added to each angle:**
  1. **+I and −I solves.** k = (T(+I) − T(−I)) / 2I, from the copper (Lorentz) torque and separately from the rotor (virtual-work) torque. Differencing cancels any magnet-only torque. The archived runs used +I only and could not remove it.
  2. **An Iamp = 0 solve.** Maxwell's magnet field of this detailed model is exported on the same layout trace centre-lines used by the field map and the analytical model. It is then integrated with the same filament torque formula.
  3. **The same export from the +I solution.** This shows whether the board's own current field changes the filament result.

How the outcome is read

If the filament torque on the detailed model's own field…	…then the gap comes from
equals the detailed copper torque	the **field**: weaker at the board in the detailed model than in the field map (mesh or region resolution)
equals the field map and the analytical value	the **conductor representation**: how current flows in the solid copper compared with centre-line paths and network current shares

The copper-against-rotor comparison separately tests whether Maxwell's force integration over the thin copper is self-consistent.

Each job ran 12 solves (8 at ±I, 4 at zero current) on 8 cores of the UCT HPC: job 1517602 (board 5) and job 1517777 (board 1), both on 22 October 2026.

Diagnostic results

At both positions the detailed model's own field reproduces the field map, and Maxwell's two force routes agree with each other. The gap is the conductor representation: +1.3 % at board 1, +9.7 % at board 5.

K, phase A (mN·m/A)	Board 1 (1.000 mm)	Board 5 (5.081 mm)
Detailed copper, +I only (archived method)	6.2818 (archive 6.2816)	1.8264 (archive 1.8266)
Detailed copper, ±I	6.2818	1.8264

K, phase A (mN·m/A)	Board 1 (1.000 mm)	Board 5 (5.081 mm)
Detailed rotor (virtual work), ±I	6.2777	1.8342
Filament on the detailed model's magnet field (Iamp = 0)	6.3618	2.0029
Filament on the detailed model's total field (+1 A)	6.3617	2.0028
FEM field map, for comparison	6.3697	2.0154
Analytical model, for comparison	6.3690	2.0054

Comparison	Board 1	Board 5	What it shows
Detailed copper, re-solved vs archive	+0.003 %	−0.01 %	The archived values are reproduced
Rotor vs copper torque, ±I	−0.07 %	+0.4 %	Maxwell's force calculation is self-consistent
Filament on detailed field vs field map	−0.1 %	−0.6 %	The detailed model's field is the same field
Filament on detailed field vs analytical	−0.1 %	−0.1 %	It also matches the independent model
Filament on detailed field vs detailed copper	+1.3 %	+9.7 %	The whole gap sits in the conductor representation
Total field vs magnet field (filament)	−0.002 %	−0.005 %	The board's own current field plays no part

The archived rotor-torque mismatch. The archived +I-only board runs showed copper and rotor torques differing by 9–60 % at single angles for boards 2–5. The diagnostic explains this. The rotor torque carries a magnet-only numerical offset: between −0.47 and +0.21 mN·m here, against a board-5 signal of about 2 mN·m. Solving at ±I cancels it. The copper torque, used for every reported K, does not carry it.

The board-1 summary file prints the verdict "MIXED". That is only because its 1.3 % difference is below the script's 3 % threshold for a clear call; the pattern is the same as at board 5, at a smaller size.

Interpretation and consequences

The detailed-copper FEM values stay the reference for K. Everything that depends only on the field is now confirmed by three independent routes.

What is established

- **The field is right.** It is the same in three routes: the detailed model's own field, the field map, and an analytical model sharing no code or data with Maxwell. They agree within 0.6 % at board 5 and 0.1 % at board 1. Results that depend on the field shape are therefore confirmed: phase angle, EMF harmonics, the S2/S1 ratio, rotating-model waveforms and steel flux density.
- **Maxwell's detailed solution is self-consistent.** Copper and rotor torque agree within 0.4 % once the magnet-only offset is cancelled.
- **The difference is in the conductor model.** The centre-line paths with DC-network current shares, used by both the field map and the analytical model, couple more strongly to the magnets than Maxwell's solid copper. The excess grows with distance: about +1.3 % at 1 mm and +9.7 % at 5 mm.

Conductor-geometry check (offline, 22 Oct): the cause located

The same copper islands that Maxwell builds (from the ODB++ export, all six layers, phase A) were rasterised at 0.1 mm and 0.05 mm. The DC current in each island was then solved as a 2D sheet, entering and leaving at the vias and terminal pads. The torque of these sheet currents was computed in the **analytical** field, so no Maxwell input is involved.

K, phase A (mN·m/A)	Board 1 (1.000 mm)	Board 5 (5.081 mm)
Analytical field, centre-line network currents	6.3749	2.0054
Analytical field, 2D sheet currents in the real copper (0.05 mm)	**6.2808**	**1.8260**
Same, 0.1 mm raster	6.2814	1.8262
FEM detailed copper (diagnostic)	6.2818	1.8264
Sheet currents vs detailed FEM	−0.02 %	−0.02 %

- With the current solved in the actual copper shapes, the independent model reproduces Maxwell's detailed copper torque to 0.02 % at both positions.

- The 1.4–10 % gap is therefore entirely the centre-line network's current routing. It assigns more current to paths that link the magnet fundamental strongly than the real copper does.
- At board 5 the excess is uniform across the six layers (−8.4 to −9.2 % per layer). At board 1 it ranges from +0.4 % on L1 to −3.3 % on L6.
- The resistance of the sheet model converges towards the FEM value: 0.302 → 0.288 Ω per board-phase as the raster is refined from 0.1 to 0.05 mm, against 0.278 Ω in the FEM and 0.264 Ω in the network (tracks only).
- Still open: which island routings in the network differ from the real copper. This only explains the network's error further; it does not change any value.

Every board, sheet currents (0.1 mm raster), analytical field, phase A (mN·m/A)

Board	Rev0 / S1: sheet model	Rev0 / S1: FEM detailed	Difference	Rev1-S S2: sheet model	Rev1-S S2: FEM detailed	Difference
B1	6.2814	6.2816	−0.00 %	6.5712	6.5593	+0.18 %
B2	4.5910	4.5905	+0.01 %	4.9482	4.9382	+0.20 %
B3	3.3654	3.3655	−0.00 %	3.8201	3.8115	+0.23 %
B4	2.4750	2.4755	−0.02 %	3.0662	3.0572	+0.29 %
B5	1.8262	1.8266	−0.02 %	2.6056	2.5965	+0.35 %
Stator	**18.539**	18.540 (full motor 18.534)	**−0.01 % (+0.03 %)**	**21.011**	20.963	**+0.23 %**

With the current solved in the real copper, the independent model predicts the machine constant:

- **Rev0:** 18.54 mN·m/A, +0.03 % against the full-motor FEM.
- **Rev1-S:** 18.54 + 21.01 = 39.55 mN·m/A, +0.13 % against the board-by-board FEM.

The phase angle at board 1 is −30.18°, identical to the detailed FEM. The small S2 residual is consistent with the different R1-steel treatment: the detailed S2 model used a finite solid plate with R1's magnets omitted, while the analytical model uses an infinite permeable plane.

Consequences for the report

Quantity	Value to report	Basis
K, Rev0 / Rev1-S	18.53 / 39.50 mN·m/A	FEM detailed copper (unchanged)
Back-EMF, harmonics, phase, S2/S1	As in the final results	FEM, now independently confirmed by the analytical model

Quantity	Value to report	Basis
Field-map K	Upper bound only, +5 % per stator	Conductor-representation bias, now explained
Analytical K	Independent check, +5.2 % / +5.45 %	Centre-line network bias; with 2D sheet currents in the real copper: 18.54 / 39.55 mN·m/A (+0.03 / +0.13 %)
Copper vs rotor torque	Agree within 0.07–0.4 % at ±I	Corrects the earlier statement in the Chapter 3 record (done 22 Oct)

No FEM value in the final results changes as a result of this work.

Resistance and operating analysis

The choice of K changes the operating torque by about 5 % but not the current limit, which depends only on resistance and cooling.

Phase resistance

Source	Per board-phase (Ω)	Per stator phase, 5 boards (Ω)
Analytical, layout (ρ = 1.724e-8 Ω·m, 0.5 mm × 35 µm tracks, vias solid or 20 µm plated)	0.266 – 0.270	1.33 – 1.35
FEM DC conduction (solid vias)	0.277 – 0.278	1.39
Measured (2-wire; temperature not recorded)	0.32 – 0.34	1.87 – 1.90, with joints and leads

The layout value is 4 % below the FEM value, which includes pads and current crowding at bends. The measured value is 20–28 % higher per board, which points to thinner finished copper or narrower etched tracks than nominal; this has not been sectioned. The operating analysis uses the **measured** resistance.

Inductance was not computed analytically. The FEM armature-only value, 34–51 µH per stator, is used and labelled FEM. Its effect on these results is under 1 %, since the reactance is about 0.04 Ω against about 2 Ω of resistance at 1000 rpm.

Operating points. Speed, current and voltage are tied together by the controller and inverter. The analysis assumes:

- star connection, FOC with i_d = 0;

- SVPWM with modulation index 0.95: 12 V dc for Rev0, 24 V dc for Rev1-S in series;
- a 60 °C board limit in 25 °C air, with copper loss at R(T) plus the FEM track eddy loss;
- thermal resistance per stator from Icepak (FEM: 8.75 / 14.8 K/W) or the cautious basis (15 / 25.3 K/W).

At 1000 rpm, cautious thermal basis	Rev0, analytical K	Rev0, FEM K	Rev1-S, analytical K	Rev1-S, FEM K
Current limit (A pk)	0.840	0.840	0.644	0.644
Electromagnetic torque (mN·m)	24.6	23.3	40.2	38.2
Line voltage at the motor (V rms)	4.75	4.63	8.79	8.52
Electromagnetic efficiency	52 %	51 %	60 %	59 %

Both machines are limited by heat up to about 2000–2400 rpm and by voltage above that. The no-load speed is 3224 / 3391 rpm for Rev0 at 12 V and 3018 / 3182 rpm for Rev1-S at 24 V (analytical / FEM K). These are electrical limits only; the mechanics stay limited to 1000 rpm. With the FEM K, the analysis reproduces the rev. D operating points.

Limitations and open items

The cause of the 5 % gap is now located in the centre-line network's current routing (see Interpretation). What remains open changes no value.

Analytical model

- **Steel:** infinite planes of infinite permeability. Finite plate radius (43 mm against a 38 mm magnet outer radius), the 19 mm bore and finite μ are neglected, and the size of that error is not quantified.
- **Magnets:** ideal uniform magnetisation. The recoil correction is linear and volume-averaged, and Br is nominal. Recoil alone shifts K by 5.3 %, and the N35 Br tolerance by up to 4.9 %.
- **Conductors:** the same centre-line network as the field map. This over-predicts K by 1.3–9.7 % per board; solving the current as 2D sheets in the real copper removes the error (0.02 %).
- **Excluded:** armature reaction (the winding field is 300–400 times weaker than the magnets'), eddy currents, and skin and proximity effects (negligible at 133 Hz).

Diagnostic

- Only phase A, at two of the five board positions. Boards 2–4 are inferred from the per-board trend, which the archived runs show as smooth.
- Four rotor angles over half an electrical period, as in the archive; at this spacing only the 7th harmonic (about 0.1 % of the fundamental) aliases onto the fundamental.
- The sheet-current check covers boards 1 and 5 directly; all ten boards of Rev0 and Rev1-S S2 were then computed the same way (Interpretation section).

Open items

1. **Conductor-geometry comparison (offline, no HPC).** Done on 22 Oct: the 2D sheet-current solve reproduces Maxwell. Remaining: identify which island routings in the centre-line network differ from the real copper.
2. **Analytical inductance**, to replace the FEM value used in the operating analysis.
3. **Measured results.** Bench testing (back-EMF, torque, heat run) remains postponed. When it happens, the measured back-EMF tests K directly against both the FEM and the analytical predictions.

Reproducibility

Every number above can be regenerated from the files and jobs listed here.

Item	Where	Notes
Analytical model, comparison, resistance, operating analysis	EEE4022S_Analytical_Verification_Phase2.zip: `analytical_em.py`, `compare.py`, `resistance.py`, `operating.py`, `sheet_test.py`	Python with numpy only; the FEM values are read only by `compare.py` (folder `fem_reference/`)
Copper paths	`traces_board.json`, built by `make_traces_board.py` from the ODB++ export	DC nodal solve, no Ansys
Diagnostic model	`diag.py`; profiles DIAG_REV0_B1 and DIAG_REV0_B5 in `config.py`	Same settings as REV0_DOE_D100_P20 / D508_P20
HPC runner	`hpc_motor.py`, sha 6a4e43101494 (HPC_more_runs folder)	Board 5 ran on the earlier build 175cab85628c, with identical model code

Item	Where	Notes
Diagnostic jobs	UCT HPC jobs 1517602 (board 5) and 1517777 (board 1), 22 Oct 2026, 8 cores each	Results in RESULTS_DIAG_REV0_B5 and RESULTS_DIAG_REV0_B1: `diag_summary.txt`, `diag_results.json`
Archived FEM references	FIELDMAP_REV0_P20, FIELDMAP_REV1S_P20, REV0_DOE_D*_P20, full-motor job 1470376	As used in the final simulation results

Run order: `python sheet_test.py`, then `python analytical_em.py`, then `python compare.py fem_reference` (about 20 minutes), then `python resistance.py` and `python operating.py`.